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Abstract

This study investigates the mechanical properties of three common infill patterns-grid, gyroid, and cubic-constructed using fused filament fabrication. Test specimens were printed with polylactic acid, and stiffness, strength, and specific energy absorption were evaluated under compression across varying relative densities. The results demonstrate the superior performance of the grid pattern in terms of stiffness and energy absorption, although this advantage is direction-dependent. The gyroid pattern exhibited competitive energy absorption with improved isotropy, while the cubic pattern showed the lowest mechanical properties across all metrics. These findings provide actionable insights for optimizing 3D-printed components for load-bearing applications by balancing material efficiency and mechanical performance.

Keywords

3d printing, general topics for engineers, infill pattern, mechanical properties

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Abstract

This study investigates the mechanical properties of three common infill patterns—grid, gyroid, and cubic—constructed using fused filament fabrication. Test specimens were printed with polylactic acid, and stiffness, strength, and specific energy absorption were evaluated under compression across varying relative densities. The results demonstrate the superior performance of the grid pattern in terms of stiffness and energy absorption, although this advantage is direction-dependent. The gyroid pattern exhibited competitive energy absorption with improved isotropy, while the cubic pattern showed the lowest mechanical properties across all metrics. These findings provide actionable insights for optimizing 3D-printed components for load-bearing applications by balancing material efficiency and mechanical performance.

Keywords: Additive Manufacturing, Infill Patterns, Mechanical Testing

1. Introduction

Additive manufacturing (AM), commonly known as 3D printing, has revolutionized modern product development by enabling the rapid and cost-effective fabrication of prototypes and end-use parts with complex geometries. AM allows for unprecedented design flexibility, offering possibilities for light-weighting and tailored mechanical properties. One of the most significant advantages of 3D printing is its ability to produce intricate internal geometries, leading to a revolution in mechanical metamaterial research. By architecting periodic unit cells, metamaterials, specifically lattice structures, can tune the properties of monolithic materials for a specific mechanical response, opening new avenues in engineering applications such as aerospace, automotive, and medical devices[1].

The most common application of lattice structures is in the standard infill patterns available in every slicing software. These patterns dictate the material distribution within a part to minimize material usage, significantly influencing the stiffness, strength, and energy absorption of the final print.

Whereas much attention is being directed to the characterization of and optimization of a wide variety of lattice structures[2,3], there remains a gap in the systematic evaluation of the most commonly used infill patterns for standard 3D printing materials. There are many infill pattern options in standard slicing software, but the relationship between infill density, pattern type, and mechanical performance remains an under-explored area in research.

This paper seeks to address this knowledge gap by investigating the specific stiffness, strength, and energy absorption of three commonly used infill patterns—grid, gyroid, and cubic—under compression. Multiple relative densities for each pattern are printed using polylactic acid (PLA), the most widely used polymer 3D printing filament. The results of this study aim to provide practical insights into the role of infill pattern and density in optimizing 3D printed components for load-bearing applications, contributing to a better

understanding of how to balance material use and mechanical performance in 3D printed structures.

2. Methods

The experimental procedure used in this study encompassed test specimen design, additive manufacturing, mass measurement, and destructive compression testing. An overview of this process is illustrated in Fig. 1.

2.1 Test Specimen Development

The test specimens used in this study had cubic geometry with edge lengths of 30 mm. The justification for cubic specimens over cylindrical ones, as is common for compression testing, is to better conform to the periodicity of the infill patterns used. Three infill patterns were investigated (grid, gyroid, and cubic) at three relative densities (10%, 20%, and 30%). These infills were encased in 0.42 mm thick walls on each of the six faces to better recreate how infill patterns are used in practice. Table 1 provides an overview of the test specimens used in this study.

To eliminate any machine-to-machine variation, all test specimens were printed simultaneously on a single build plate. To achieve this, the Bambu Studio slicer was used to vary the infill type and relative density of all 27 samples. A Bambu Carbon X1 (Bambulab USA Inc, Austin, Texas) was used to print the test specimens with Bambu Lab's purple PLA Basic filament. The slicing and printing process is shown in Fig. 1a and Fig. 1b, respectively.

Table 1 Test Specimens

Infill Pattern	Density, %	Dimensions, mm	Trials
Grid	10, 20, 30	30×30×30	3
Gyroid	10, 20, 30	30×30×30	3
Cubic	10, 20, 30	30×30×30	3

2.2 Mechanical Testing

To ascertain the specific energy absorption of each specimen, defined as the area under the load-displacement curve divided by the mass of the sample[4], they were weighed by a Mettler Toledo Precision Balance (Mettler Toledo, Columbus, Ohio), as shown in Fig. 1d.

Compression testing was performed on a universal testing machine (Instron, Norwood, Massachusetts), with a strain rate of 0.005/s up to a maximum strain of 75%.

2.3 Data Analysis

After experimental testing, data was post-processed

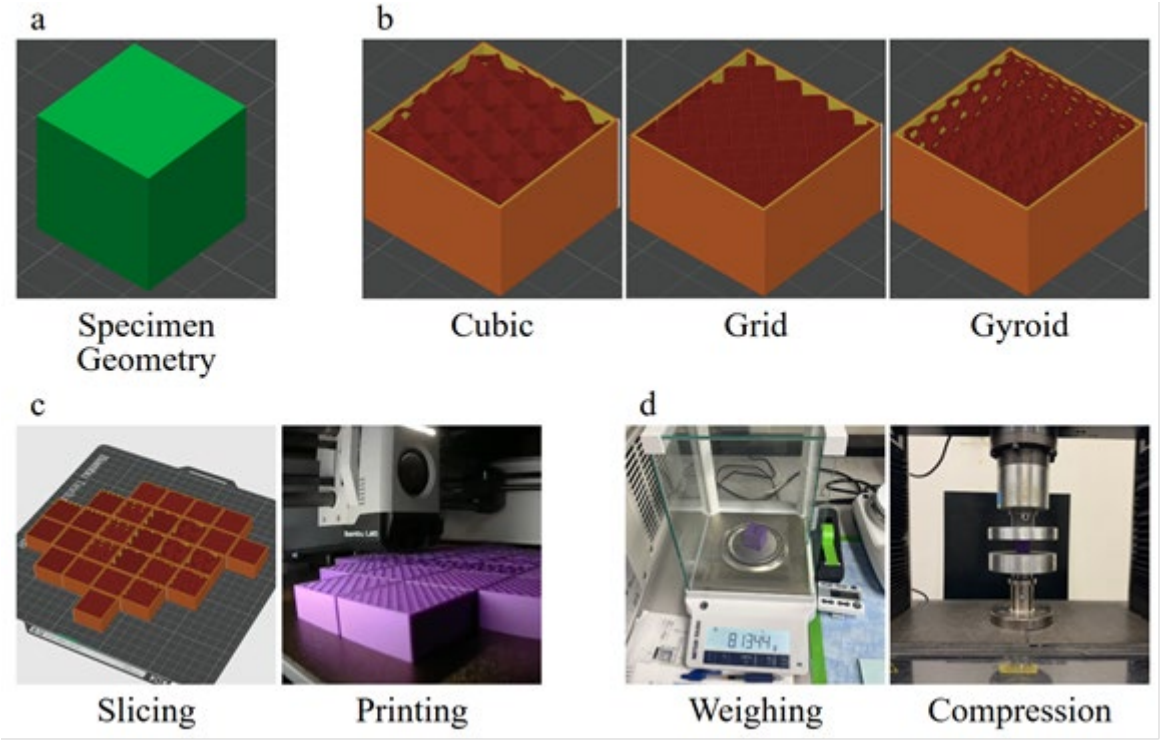


Fig. 1 Experimental Setup

(a) Specimen geometry. (b) Infill pattern types. (c) Additive manufacturing process. (d) Experimentation.

and analyzed with Matlab (MathWorks, Natick, Massachusetts). Plots of load versus displacement were defined and integrated to compute energy absorption. Stiffness was determined from the elastic region of the load versus displacement graph. For each elastic region, iteration was used to identify the segment of displacement that produced a coefficient of determination closest to one. For the quantities of interest, standard errors were computed and an analysis of variance was performed.

3. Results

The average mass and standard error of the printed specimens are reported in Table 2.

Table 2 Specimen mass measurements.

Infill Pattern	Density, %	Average Mass, g	Std Dev, g
Grid	10	8.1162	0.0174
	20	10.4049	0.0045
	30	13.2457	0.0297
Gyroid	10	7.7805	0.0904
	20	10.3857	0.0174
	30	12.8249	0.0150
Cubic	10	7.7483	0.0091
	20	10.4397	0.0306
	30	13.0415	0.0291

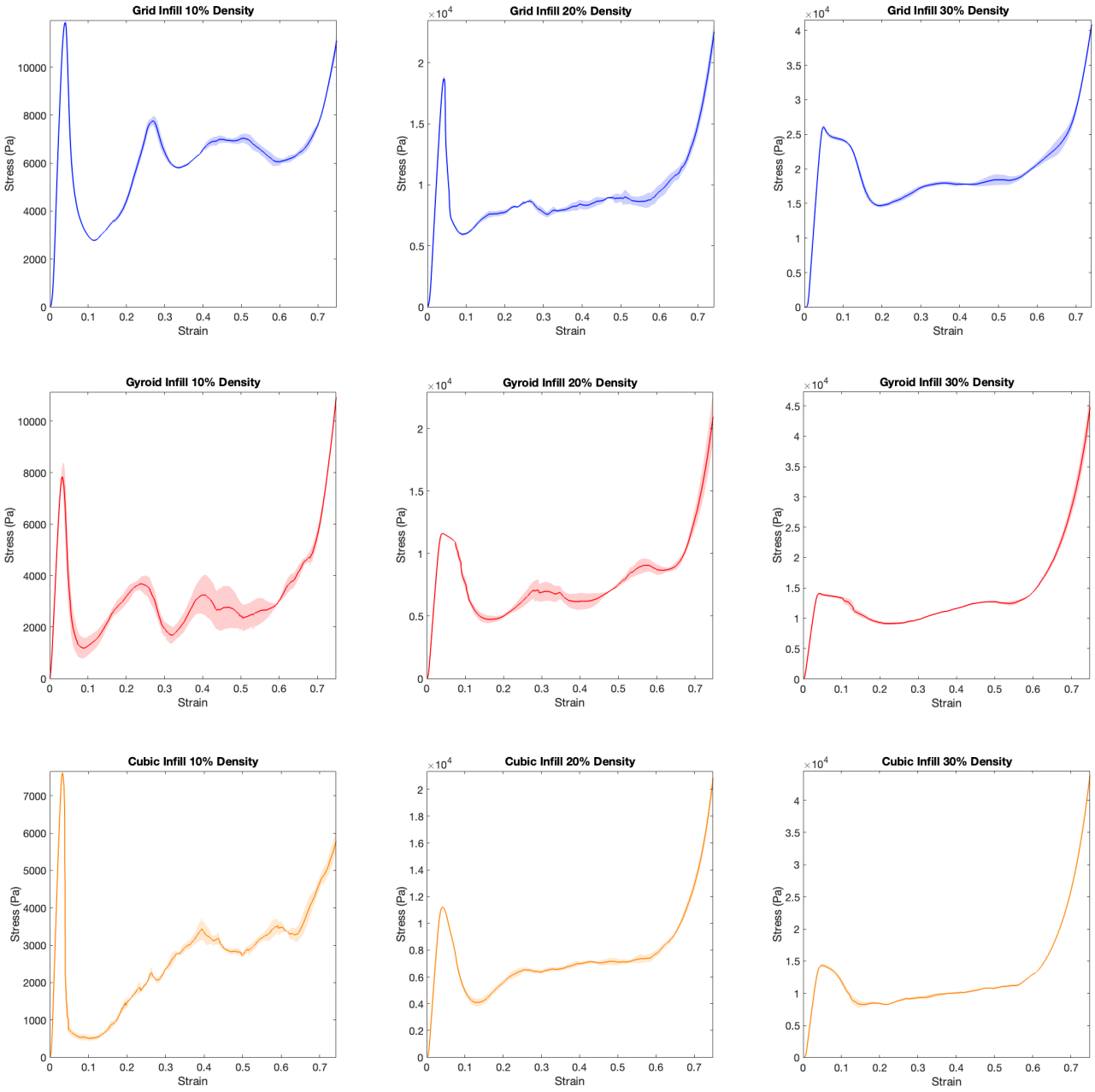


Fig. 2 Average stress-strain response of each infill pattern and relative density.

The standard deviation amongst the three trials of each specimen configuration is very low relative to average mass. While this demonstrates the precision of the manufacturing process, it also reveals how each infill pattern uses a different amount of material despite possessing the same infill density.

Through compressive testing on the universal testing machine, the load-displacement response of all 27 test specimens was recorded. The stress-strain relationship for each specimen was then determined by normalizing load and displacement measurements by the standardized cross-sectional area (0.0009 m^2) and side length (30 mm), respectively.

For each infill type and density, three trials were performed and the stress-strain responses of the three corresponding test specimens were averaged to form an aggregate relationship. The standard error was calculated by determining the standard deviation of stress at each value of strain and dividing it by the square root of the sample size. These trends are visually depicted in Fig. 2, where the bold lines represent the averaged responses, and the shaded areas illustrate the standard error. The minimal extent of these colored regions highlights the printer's consistency, as specimens of the same type exhibited nearly identical stress-strain behavior.

The relationship between relative density and standard error was visually observed to varied arbitrarily by infill type. Interestingly, in comparison to the other two infill types, cubic appears to demonstrated superior reproducibility with negligible variation throughout all three relative densities.

Additional plots were used to enable broader comparisons. Fig. 3 organizes the data by infill pattern, highlighting performance variations at each density when the infill type is fixed. In contrast, Fig. 4 shows the average stress-strain responses of the test specimens grouped by relative density, illustrating how this quality influences mechanical performance across all infill patterns. Together, these graphs provide complementary insights into the roles of density and infill pattern in determining stress-strain behavior.

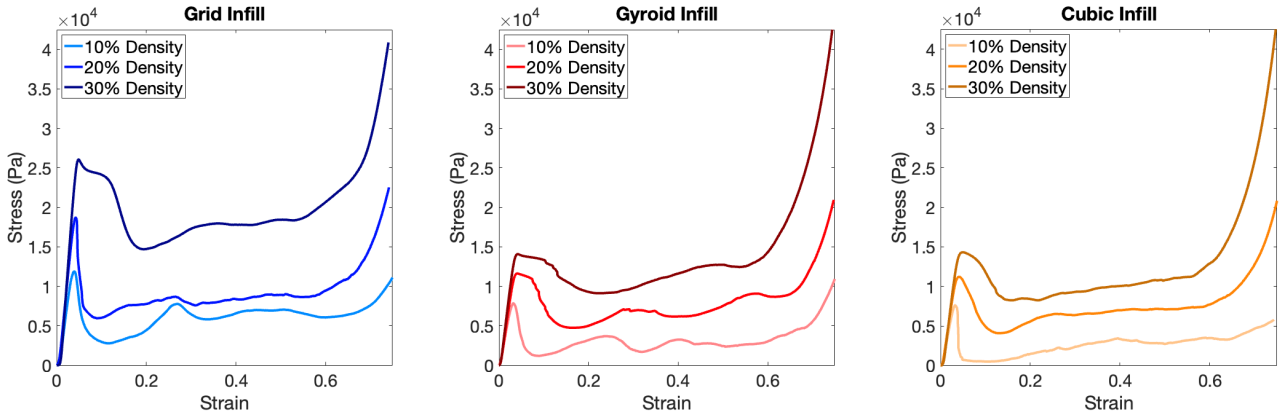


Fig. 3 Comparison of average stress-strain responses at each relative density of each infill type.

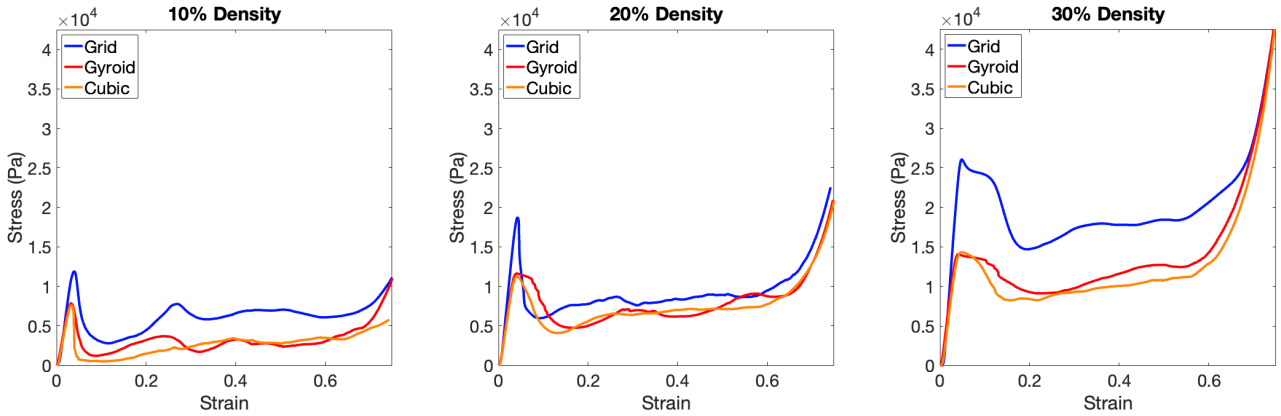


Fig. 4 Comparison of average stress-strain responses of each infill type at each relative density.

4. Discussion

The results provide a quantitative basis for assessing the trade-offs between stiffness, SEA, and density. Both SEA and stiffness increase with relative density across all patterns, reflecting the fundamental trade-off between material efficiency and mechanical performance. At lower densities, performance differences between patterns are most pronounced, likely due to the greater influence of geometric imperfections and print quality. At higher densities, the differences narrow, particularly between the grid and gyroid patterns, emphasizing their suitability for applications requiring enhanced mechanical properties.

The practical implications of these findings are significant. The grid pattern's superior stiffness and SEA make it ideal for structural supports and load-bearing components in predictable loading scenarios, such as frames or beams. The gyroid pattern, with its isotropic geometry, excels in applications requiring uniform load distribution and multi-directional performance. For instance, it could enhance crash-resistant structures, such as automotive crumple zones, by dissipating energy evenly across multiple directions. Similarly, its isotropy makes it a promising candidate for orthopedic scaffolds, where uniform stress distribution is critical for promoting tissue integration and long-term durability. Lastly, the relatively low variation in the stress-strain response of the cubic infill pattern has advantages in mass-produced products to maintain consistent performance.

Despite the study's strengths, several limitations must be acknowledged. The focus on compressive loading along a single axis constrains the breadth of the findings, particularly for applications involving multi-directional forces. Multi-directional load testing would likely reveal weaknesses in the grid pattern's anisotropy and further highlight the gyroid pattern's advantages in isotropic performance. Moreover, exploring additional polymers and alternative infill patterns would provide a deeper understanding of the mechanical properties of 3D-printed structures.

4.1 Standard Error

To quantify the consistency of stress-strain response for each specimen tested, we calculated the average standard error of stress across each strain value measured, as shown in Fig. 5. Although these plots disregard the state of the material at each relative density, the mean value gives a general overview of the consistency of each infill pattern. As shown in Fig. 5, the mean standard error of grid infill specimens strictly increased with increasing relative density from 157.17 Pa to 359.67 Pa and finally to 416.39 Pa. On the other hand, the mean standard error of gyroid infill specimens hovered at roughly the same value for each relative density, oscillating from 398.48 Pa to 439.51 Pa and finally to 344.83 Pa. Most notably, it was confirmed that cubic infill specimens demonstrated the least variation throughout, achieving the lowest mean standard error for every relative density. This analysis demonstrates that cubic infill was the most consistent and reproducible infill type by the Bambu Carbon X1.

4.2 Specific Energy Absorption

Specific energy absorption (SEA), shown in Fig. 6, reveals nuanced performance characteristics across different infill patterns. At lower densities (10%), the grid pattern demonstrates a comparably high SEA, achieving an SEA of $(0.3145 \pm 0.0396) \times 10^6$ J/kg. The gyroid and cubic patterns underperform at this density, with SEAs of $(0.1434 \pm 0.0248) \times 10^6$ J/kg and $(0.1787 \pm 0.0414) \times 10^6$ J/kg, respectively. As density increases

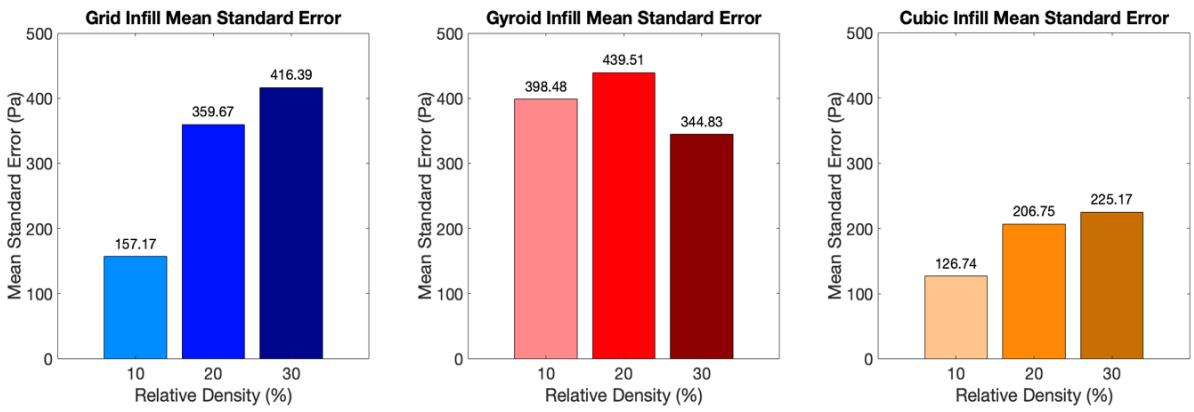


Fig. 5 Comparison of mean standard error of stress for each infill type and relative density.

to 20%, the gyroid pattern outperforms the grid pattern, reaching an SEA of (0.4646 ± 0.0248) × 10^6 J/kg, compared to the grid's (0.4588 ± 0.0972) × 10^6 J/kg. At 30% density, the grid pattern exhibits the highest SEA, achieving (1.0215 ± 0.0482) × 10^6 J/kg, followed by the gyroid pattern with an SEA of (0.6928 ± 0.0179) × 10^6 J/kg and the cubic pattern remaining slightly lower at (0.6286 ± 0.0196) × 10^6 J/kg. The data suggests that the choice of material and infill density can significantly impact the specific energy absorption properties. These results highlight that all patterns show a growth in SEA as densities get higher. Overall, the grid pattern is observed to perform the best across all the density values.

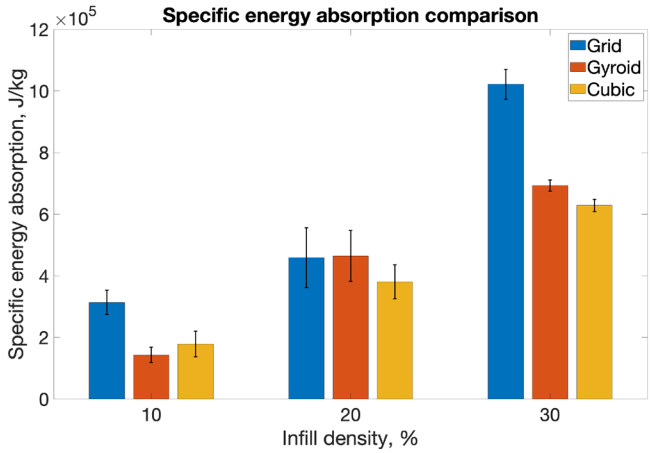


Fig. 6 Specific energy absorption versus relative density of each infill type.

4.3 Stiffness

The stiffness results, as illustrated in Fig. 7, parallel the SEA trends. At 10% density, the grid pattern achieves a stiffness of (0.0113 ± 0.0004) × 10^6 N/m, outperforming both the gyroid (0.0056 ± 0.0030) × 10^6 N/m and cubic patterns (0.0046 ± 0.0002) × 10^6 N/m. As density increases, the grid pattern maintains its lead, with stiffness values reaching (0.0123 ± 0.0007) × 10^6 N/m at 20% while the gyroid pattern also improves with density as well as the cubic pattern, achieving (0.0210 ± 0.0009) × 10^6 N/m, and (0.0099 ± 0.0008) × 10^6 N/m respectively, remaining slightly behind the grid pattern. As the density goes up to 30%, the grid pattern once again overperforms the gyroid pattern and the cubic pattern, showing the stiffness value of (0.0210 ± 0.0009) × 10^6 N/m, (0.0137 ± 0.0005) × 10^6 N/m, and (0.0120 ± 0.0006) × 10^6 N/m respectively. These results reinforce the grid pattern's effectiveness in resisting deformation along the loading axis while emphasizing the gyroid pattern's balanced performance for more variable applications. However, it is important to note that the large standard deviation for the gyroid pattern at 10% density makes it difficult to conclude much about its stiffness at this lower density.

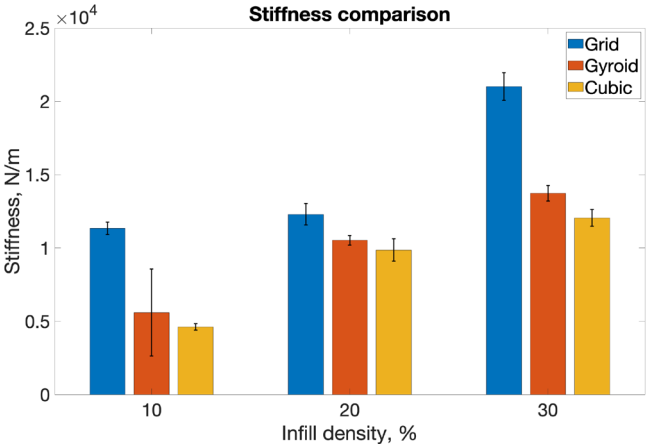


Fig. 7 Stiffness versus relative density of each infill type.

4.3 Yield Strength

The yield strength results, illustrated in Fig. 8, reveal a notable performance distinction among the three infill types. Grid consistently exhibited the highest yield strength across all relative densities, emphasizing its superior structural integrity. At 10% infill density, Grid achieved a yield strength of 11.87 MPa, outperforming Gyroid (7.85 MPa) and Cubic (7.61 MPa) by significant margins. This trend continued at 20% density, with Grid yielding 18.70 MPa compared to Gyroid's 11.62 MPa and Cubic's 11.21 MPa. At the highest density of 30%, Grid further cemented its dominance with a yield strength of 26.06 MPa, nearly double that of Gyroid (14.08 MPa) and Cubic (14.32 MPa). Notably, as the relative infill density increased, the difference in yield strength between grid and the other two infill types became more pronounced. At 30%, the grid infill pattern demonstrated a yield strength that was almost double of that of the other two infill types. This is interesting because one might typically expect infill type to be less of a factor as the relative density increases, approaching a monolithic material.

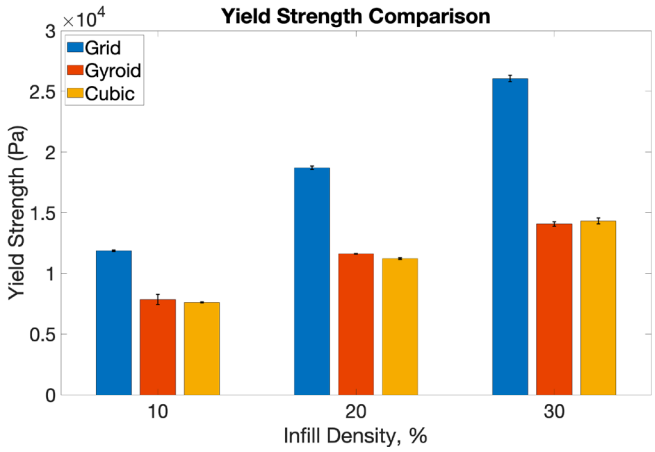


Fig. 8 Yield strength versus relative density of each infill type.

4.2 Analysis of Variance

An analysis of variance (ANOVA) was also performed to assess the statistical significance of the influence of relative density on the stiffness (Table 3) and specific energy absorption (Table 4) across the three infill patterns. This analysis provides insight into whether variations in relative density lead to measurable differences in these mechanical properties and how these effects differ between the grid, gyroid, and cubic infill structures.

Table 3 ANOVA for Stiffness.

Infill Pattern	F-Value	p-Value
Grid	377.13	4.9154×10^{-7}
Gyroid	354.33	5.9175×10^{-7}
Cubic	356.9	5.7919×10^{-7}

Table 4 ANOVA for Specific Energy Absorption.

Infill Pattern	F-Value	p-Value
Grid	640.17	1.0148×10^{-7}
Gyroid	65.597	8.3646×10^{-5}
Cubic	549.34	1.6023×10^{-7}

It is apparent that stiffness is strongly influenced by relative density, with the grid pattern having the strongest dependence. For the grid and cubic patterns, specific energy absorption is also heavily influenced by relative density. The F-value of the gyroid pattern is substantially smaller than the other two infills for specific energy absorption. This is most likely due to the gyroid possessing the highest variation of stress-strain response of all the samples tested. Nevertheless, the specific energy absorption of the gyroid pattern is still heavily influenced by relative density.

5. Conclusion

This study provides an evaluation of the mechanical properties of three common 3D printing infill patterns—grid, gyroid, and cubic—under compressive loading at multiple relative densities. By systematically analyzing specific energy absorption and stiffness, this work elucidates the trade-offs inherent in selecting infill patterns for additive manufacturing applications.

The results demonstrate that the grid pattern consistently achieves superior mechanical performance, with the highest SEA and stiffness across all densities. These qualities make it an excellent choice for load-bearing applications requiring high directional strength and energy dissipation, such as structural supports and beams.

The gyroid pattern, while slightly underperforming compared to the grid in stiffness and SEA, offers significant advantages in isotropic performance due to its uniform geometry. This makes it ideal for applications involving multi-directional loading or energy absorption, such as crash-resistant structures and biomedical scaffolds.

The study also highlights the strong influence of relative density on both specific energy absorption and stiffness for all three infill patterns. As density increases, the mechanical performance improves, with the grid and gyroid patterns narrowing in performance differences at higher densities. The results emphasize the importance of carefully selecting infill density and pattern to balance material usage with mechanical requirements.

While the findings provide valuable insights, the study acknowledges its limitations, including the focus on uniaxial compressive loading and the exclusive use of polylactic acid (PLA) as the test material. Future research should explore multi-directional loading scenarios, alternative materials, and hybrid infill designs to further refine our understanding of 3D-printed structures.

Ultimately, this work contributes to the growing body of knowledge in additive manufacturing. These insights enable engineers and designers to better tailor 3D-printed components for specific applications, paving the way for more efficient and effective use of this transformative technology.

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Conflict of Interest

The authors declare no conflict of interest.

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